

Summer 1979

Problem 1 (Su79, Fa93). Prove that the matrix

$$\begin{pmatrix} 0 & 5 & 1 & 0 \\ 5 & 0 & 5 & 0 \\ 1 & 5 & 0 & 5 \\ 0 & 0 & 5 & 0 \end{pmatrix}$$

has two positive and two negative eigenvalues (counting multiplicities).

Problem 2 (Su79). Let $\mathbb{F}$ be a subfield of a field $\mathbb{K}$, and $p, q \in \mathbb{F}[x]$. Prove that their greatest common divisor in $\mathbb{F}[x]$ is the same as their gcd in $\mathbb{K}[x]$.

Problem 3 (Su79). Let X be the space of orthogonal real $n \times n$ matrices. Let $v_0 \in \mathbb{R}^n$. Locate and describe the elements of X , where the map

$$f : X \rightarrow \mathbb{R} \quad f(A) = \langle v_0, Av_0 \rangle$$

takes its maximum and minimum values.

Problem 4 (Su79). Prove that the group of automorphisms of a cyclic group of prime order p is cyclic and find its order.

Problem 5 (Su79, Fa97). 1. Give an example of a differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ whose derivative f' is not continuous.

2. Let f be as in Part 1. If $f'(0) < 2 < f'(1)$, prove that $f'(x) = 2$ for some $x \in [0, 1]$.

Problem 6 (Fa78, Su79). Let $f(z) = a_0 + a_1z + \cdots + a_nz^n$ be a complex polynomial of degree $n > 0$. Prove

$$\frac{1}{2\pi i} \int_{|z|=R} z^{n-1} |f(z)|^2 dz = a_0 \bar{a}_n R^{2n}$$

Problem 7 (Su79, Sp82, Sp91, Sp96). Let f be a continuous complex valued function on $[0, 1]$, and define the function g by

$$g(z) = \int_0^1 f(t) e^{tz} dt \quad z \in \mathbb{C}$$

Prove that g is analytic in the entire complex plane.

Problem 8 (Su79). Let $U \subset \mathbb{R}^n$ be an open set. Suppose that the map $h : U \rightarrow \mathbb{R}^n$ is a homeomorphism from U onto $\mathbb{R}^n$, which is uniformly continuous. Prove $U = \mathbb{R}^n$.

Problem 9 (Su79, Fa96). Prove that a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ has

1. a one-dimensional invariant subspace, and
2. a two-dimensional invariant subspace.

Problem 10 (Su79, Fa79, Fa82, Su85). Find all pairs of C^∞ functions $x(t)$ and $y(t)$ on $\mathbb{R}$ satisfying

$$x'(t) = 2x(t) - y(t) \quad y'(t) = x(t)$$

Problem 11 (Su79, Fa82). Let A and B be $n \times n$ matrices over a field $\mathbb{F}$ such that $A^2 = A$ and $B^2 = B$. Suppose that A and B have the same rank. Prove that A and B are similar.

Problem 12 (Su79). Which rational numbers t are such that

$$3t^3 + 10t^2 - 3t$$

is an integer?

Problem 13 (Su79, Sp82, Sp83, Su84, Fa91, Fa98, Sp11, Fa15, Fa16, Fa23). Let $\mathbb{F}$ be a finite field with q elements and let V be an n -dimensional vector space over $\mathbb{F}$.

1. Determine the number of elements in V .
2. Let $\text{GL}_n(\mathbb{F})$ denote the group of all $n \times n$ nonsingular matrices over $\mathbb{F}$. Determine the order of $\text{GL}_n(\mathbb{F})$.
3. Let $\text{SL}_n(\mathbb{F})$ denote the subgroup of $\text{GL}_n(\mathbb{F})$ consisting of matrices with determinant 1. Find the order of $\text{SL}_n(\mathbb{F})$.

Problem 14 (Su79, Fa93). Let A , B , and C be finite abelian groups such that $A \times B$ and $A \times C$ are isomorphic. Prove that B and C are isomorphic.

Problem 15 (Su79). Show that $\sum_{n=0}^{\infty} \frac{z}{(1+z^2)^n}$ converges for all complex numbers z exterior to the lemniscate $|1+z^2|=1$.

Problem 16 (Su79). Let $g_n(z)$ be an entire function having only real zeros, $n = 1, 2, \dots$. Suppose

$$\lim_{n \rightarrow \infty} g_n(z) = g(z)$$

uniformly on compact sets in $\mathbb{C}$, with g not identically zero. Prove that $g(z)$ has only real zeros.

Problem 17 (Su79). Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be such that

$$f^{-1}(0) = \{v \in \mathbb{R}^3 \mid \|v\| = 1\}$$

Suppose f has continuous partial derivatives of orders ≤ 2 . Is there a $p \in \mathbb{R}^3$ with $\|p\| \leq 1$ such that

$$\frac{\partial^2 f}{\partial x^2}(p) + \frac{\partial^2 f}{\partial y^2}(p) + \frac{\partial^2 f}{\partial z^2}(p) \geq 0?$$

Problem 18 (Su79). Let E be a three-dimensional vector space over $\mathbb{Q}$. Suppose $T : E \rightarrow E$ is a linear transformation and $Tx = y$, $Ty = z$, $Tz = x + y$, for certain $x, y, z \in E$, $x \neq 0$. Prove that x , y , and z are linearly independent.

Problem 19 (Su79). Let $\{f_n\}$ be a sequence of continuous real functions defined $[0, 1]$ such that

$$\int_0^1 (f_n(y))^2 dy \leq 5$$

for all n . Define $g_n : [0, 1] \rightarrow \mathbb{R}$ by

$$g_n(x) = \int_0^1 \sqrt{x+y} f_n(y) dy$$

1. Find a constant $K \geq 0$ such that $|g_n(x)| \leq K$ for all n .
2. Prove that a subsequence of the sequence $\{g_n\}$ converges uniformly.

Problem 20 (Su79). Let $x : \mathbb{R} \rightarrow \mathbb{R}$ be a solution to the differential equation

$$5x'' + 10x' + 6x = 0$$

Prove that the function $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$f(t) = \frac{x(t)^2}{1 + x(t)^4}$$

attains a maximum value.